FITRI MAHARANI

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SUMMARY

Telecommunication Engineering graduate and AI/ML Engineer learner at the Apple Developer Institute for AIML, focused on building AI that feels natural and useful in everyday life. Experienced in applied AI across Machine Learning, Computer Vision, Speech Processing, LLMs, and RAG, taking models from data preparation and training all the way into native iOS apps. Brings hands-on network deployment and monitoring experience from an industry internship.

EXPERIENCE

AI/ML Engineer **April 2026 – Present**
Apple Developer Institute for AIML **Surabaya**

- Develop machine learning models in a project-based AI/ML program, working in small teams with mentor guidance to take each idea from raw data to a working model.
- Create end-to-end AI projects covering computer vision, speech recognition, LLMs and RAG, and predictive modeling.
- Run the full ML pipeline, from data collection and preprocessing to model training, evaluation, and on-device deployment with Core ML, ONNX, and MLX.

Omah IoT **Feb 2024 – July 2024**
Hardware Development Intern – Polines Data Center **Semarang**

- Deployed and configured LAN and wireless networks, including topology design.
- Installed and centrally managed Ruckus access points.
- Monitored network performance with PRTG and ran testing and troubleshooting to keep connectivity stable, secure, and reliable.

PROJECTS

Littlebrain – On-Device Flashcards From What You Read
Foundation Models, Translation, SwiftData, App Intents, WidgetKit, SwiftUI

- Built an iPhone app that turns articles, AI answers, and screenshots into flashcards, then brings them back on a five-box Leitner schedule through interactive notifications and a widget.
- Generated every card with Apple's on-device Foundation Models, routed Indonesian input through the Translation framework, and enforced card rules in a Swift verifier instead of the prompt, backed by 406 unit tests.

U See – Multi-Camera Movement Analytics
YOLO11s, BoT-SORT, OSNet Re-ID, Homography, Qwen3-8B, SwiftUI

- Built a macOS app with a team of three that turns multi-camera CCTV footage into movement paths, heatmaps, and zone analytics on a floor plan, with input from Ciputra University's Department of Architecture.
- Owned the ML pipeline: person detection with YOLO11s, tracking with BoT-SORT, and cross-camera re-identification with OSNet, plus an on-device Qwen3-8B chatbot that answers questions about each analysis.

Pezen – Voice-Driven Food Ordering
SwiftUI, SileroVAD, Speech-to-Text, Text-to-Speech, Python

- Built an iPadOS app that takes restaurant orders through natural spoken conversation in real time.
- Designed the full voice pipeline: end-of-speech detection with SileroVAD, transcription, disfluency repair, intent extraction, validation against the live menu, and spoken response.

LearnToRecall – Textbook Study Assistant
RAG, ChromaDB, BM25, Gemma 3 4B, Qwen2.5-VL, MLX, SwiftUI

- Built a RAG system with hybrid retrieval (ChromaDB + BM25) and Reciprocal Rank Fusion, keeping every

answer grounded in the user's own material.
- Ran Gemma 3 4B locally through MLX, so no user data ever leaves the device.

Receipt Scanner – Restoring and Reading Damaged Receipts
NAFNet, PaddleOCR, ONNX, FastAPI, PyTorch, SwiftUI
- Trained a NAFNet image restoration model (23.8M parameters) on 4,445 image pairs to reconstruct damaged and blurred Indonesian retail receipts.
- Combined PaddleOCR served over FastAPI with layered regex parsing to extract totals, and deployed the restoration model on device via ONNX.

House Price Predictor – Price Estimates Across 14 Cities
XGBoost, Core ML, SwiftUI, Python, Web Scraping
- Scraped 12,000+ property listings and cleaned them down to 4,206 training records.
- Trained an XGBoost model with manual hyperparameter tuning using only 5 features, then converted it to Core ML for on-device inference.

Hydroponic Lettuce Disease Detection - Undergraduate Thesis
CNN, Raspberry Pi 4, Camera, IoT, Telegram Bot
- Built an IoT system that photographs hydroponic lettuce with a Raspberry Pi 4 camera and uses a CNN to classify each plant as healthy, wilted, nutrient-deficient, or stressed by excess pH.
- Reached 96.88% accuracy on test data and 88.44% in real-world deployment, with 0% data loss in transmission and automatic Telegram alerts delivered in about 2 seconds.

Edge-to-Cloud Data Sync for IoT Hydroponic - Research Project
ESP32 (TTGO T-Call SIM800L), IoT, Edge Computing, Cloud
- Contributed to a research team building a system that keeps hydroponic sensor data in sync between edge devices and the cloud, so monitoring keeps working when the internet connection drops.
- Ran an 8-day field test with no data loss and an average transmission delay of 3 minutes. Published in IJASEIT, Vol. 15, No. 2, 2025.

EDUCATION

Bachelor of Applied Science in Telecommunication Engineering **Oct 2021 – Sep 2025**
GPA 3.71 – Semarang State Polytechnic

- Core telecom coursework: Mobile Communication Systems, Satellite and Terrestrial Communications, Fiber Optics Communication, Antenna and Propagation, High Frequency and Microwave Engineering, and Broadcasting Technique.
- Networking and hardware track: Data Communication, Computer Networks, Advanced Computer Networks I–II, Computer Network Design, Embedded Systems, and Microprocessor Interfacing.
- Two semesters of Digital Signal Processing alongside Probability and Statistics and Numerical Methods, a foundation that carries over into speech and computer vision work.

State Senior High School 3 Semarang **2018 - 2021**

ORGANIZATION

Polines TV – Treasurer **2022 – 2024**

- Managed and oversaw the organization's finances with full accuracy and transparency.
- Prepared and reported production project budgets (RAB) for Polines TV.

LICENSES & CERTIFICATIONS

- Cisco Certified Network Associate: Enterprise Networking, Security, and Automation
- Cisco Certified Network Associate version 7: Switching, Routing, and Wireless Essentials

- AWS Cloud Practitioner Essentials

LANGUAGES

- Indonesia: Native or bilingual proficiency
- English: Upper-intermediate proficiency

SKILLS

Soft Skills
- Communication: Explains technical ideas clearly, in writing and in presentations.
- Collaboration : Works well in small project teams and stays open to feedback.
- Problem Solving : Breaks big problems into small steps and tests each fix.
- Adaptability : Picks up new tools and fields quickly, from networking to AI.
- Time Management : Plans around deadlines and keeps several tasks on track.
- Attention to Detail : Double-checks work to keep results accurate.
- Curiosity : Enjoys exploring new technology and learning by building.

Hard Skills
- AI/ML : Machine Learning, Computer Vision, Speech Processing, LLMs, RAG
- Frameworks & Tools : PyTorch, TensorFlow, XGBoost, MLX, ONNX, FastAPI, ChromaDB
- Programming : Python, Swift, C
- Apple Platforms : SwiftUI, Core ML, Foundation Models, SwiftData, Xcode
- Networking & IoT : LAN/WLAN deployment, Cisco routing and switching, Ruckus, PRTG, Raspberry Pi
- Cloud : AWS and Huawei Cloud fundamental